# AQ1.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.

**5.1 Level 1:**

*1 point*

Choose the set of correct options

- [ ] If $A$A is a real $3 \times 33 \times 3$ matrix, then $det(A) = det(A^T)$det(A) = det(AT ).
- [ ] If $A = [a_{ij}]$A = [aij ] is a real $4 \times 44 \times 4$ matrix, then the order of the sub matrix obtained by deleting the $i$i-th row and $j$j-th column of $A$A is $4 \times 44 \times 4$.
- [ ] If $A = [a_{ij}]$A = [aij ] is a real $4 \times 44 \times 4$ matrix, then the order of the sub matrix obtained by deleting the $i$i-th row and $j$j-th column of $A$A is $3 \times 33 \times 3$.
- [ ] If $A$A is a real $3 \times 33 \times 3$ matrix, then the orders of all possible square sub matrices of $A$A are $1 \times 1, 2 \times 2,$ and $3 \times 31 \times 1, 2 \times 2,$ and $3 \times 3$.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If $A$A is a real $3 \times 33 \times 3$ matrix, then $det(A) = det(A^T)$det(A) = det(AT ).

If $A = [a_{ij}]$A = [aij ] is a real $4 \times 44 \times 4$ matrix, then the order of the sub matrix obtained by deleting the $i$i-th row and $j$j-th column of $A$A is $3 \times 33 \times 3$.

If $A$A is a real $3 \times 33 \times 3$ matrix, then the orders of all possible square sub matrices of $A$A are $1 \times 1, 2 \times 2,$ and $3 \times 31 \times 1, 2 \times 2,$ and $3 \times 3$.

*1 point*

Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$A = [322 232 211] be a $3 \times 33 \times 3$ matrix. Which of the following is(are) correct?

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- [ ] $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) − 2 × det ([2221 ]) + 2 × det ([2231])
- [ ] $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$det(A) = 3 × det ([3121]) + 2 × det ([2122 ]) + 2 × det ([2231])
- [ ] $det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = −2 × det ([2121]) + 3 × det ([3221 ]) − 2 × det ([3221])
- [ ] $det(A) = 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) - 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$det(A) = 2 × det ([2121]) − 3 × det ([

]) + 2 × det ([3221])

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$det(A) = 3 \times det([\begin{smallmatrix}3 & 2\\ 1 & 1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}2 & 2\\ 2 & 1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2 & 3\\ 2 & 1\end{smallmatrix}])$det(A) = 3 × det ([3121]) − 2 × det ([2221

]) + 2 × det ([2231])

$det(A) = 3 \times det([\begin{smallmatrix}3 & 2\\ 1 & 1\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2 & 2\\ 1 & 2\end{smallmatrix}]) + 2 \times det([\begin{smallmatrix}2 & 3\\ 2 & 1\end{smallmatrix}])$det(A) = 3 × det ([3121]) + 2 × det ([2122

]) + 2 × det ([2231])

$det(A) = -2 \times det([\begin{smallmatrix}2 & 2\\ 1 & 1\end{smallmatrix}]) + 3 \times det([\begin{smallmatrix}3 & 2\\ 2 & 1\end{smallmatrix}]) - 2 \times det([\begin{smallmatrix}3 & 2\\ 2 & 1\end{smallmatrix}])$det(A) = −2 × det ([2121]) + 3 × det ([3221

]) − 2 × det ([3221])

*1 point*
If all the elements of a $3 \times 3$3 × 3 real matrix $A$A are the same, then which of the following is (are) correct?

- [ ] Determinant of matrix $A$A is 00.
- [ ] Determinant of matrix $A$A cannot be determined from the given information.
- [ ] Determinant of matrix $A$A will be the sum of the elements of a row.
- [ ] Determinant of matrix $A + A^T$A + AT is 00, where $A^T$AT denotes the transpose of $A$A.
- [ ] Determinant of matrix $A + A^T$A + AT cannot be determined from the given information, where $A^T$AT denotes the transpose of $A$A.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Determinant of matrix $A$A is 00.
Determinant of matrix $A + A^T$A + AT is 00, where $A^T$AT denotes the transpose of $A$A.
*1 point*
Choose the set of correct options

- [ ] If two rows are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.
- [ ] If two columns are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.
- [ ] If one row is a non-zero multiple of another row in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is not zero.
- [ ] If one column is a non-zero multiple of another column in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
If two rows are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.
If two columns are the same in a $3 \times 3$3 × 3 real matrix, then the determinant of that matrix is zero.
If one column is a non-zero multiple of another column in a $3 \times 3$3 × 3 real matrix, then the determinant

of that matrix is zero.

Let $A = \begin{bmatrix} 1010 \\ 0101 \\ 1001 \\ 0001 \end{bmatrix}$A = $\begin{bmatrix} \\ \\ \\ 1010010010000111 \end{bmatrix}$ be a matrix of order $4 \times 4$4 × 4. Find $det(A)$det(A).

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) -1
*1 point*

**5.2 Level 2:**

*1 point*
Let $A, B,$ A, B, and $C$C be $3 \times 3$3 × 3 real matrices. Which of the following options is (are) correct?

- [ ] $det(ABC) = det(A)det(B)det(C)$det(ABC) = det(A)det(B)det(C)
- [ ] $det(A^3) = det(A)^3$ det(A3) = det(A)3
- [ ] $det(A + B + C) = det(A) + det(B) + det(C)$det(A + B + C) = det(A) + det(B) + det(C)
- [ ] $det(AB^T) = det(A)\ det(B)$det(ABT ) = det(A) det(B), where $B^T$ BT denotes the transpose of $B$ B.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$det(ABC) = det(A)det(B)det(C)$det(ABC) = det(A)det(B)det(C)
$det(A^3) = det(A)^3$ det(A3) = det(A)3
$det(AB^T) = det(A)\ det(B)$det(ABT ) = det(A) det(B), where $B^T$ BT denotes the transpose of $B$ B.
*1 point*
Let $A = [\delta_{ij}]$A = [δij] be a matrix of order $3 \times 3$3 × 3 such that $\delta_{ij} = \begin{cases} 0 \text{if } i > j \\ j \text{if } i \leq j \end{cases}$δij = {0jif i > jif i ≤ j

Choose the set of correct options

[Hint: Write down matrix $A$A first.]

- [ ] $A$A is an upper triangular matrix.
- [ ] $A$A is a lower triangular matrix.
- [ ] $det(A) = 6$det(A) = 6
- [ ] $det(A) = 3$det(A) = 3

**No, the answer is incorrect.**
**Score: 0**

**Accepted Answers:**
$A$A is an upper triangular matrix.
$det(A)$ = 6det(A) = 6

*1 point*

Let $A$ = $[\delta_{ij}]$A = [δij] be a matrix of order $3 \times 3$3 × 3 such that $\delta_{ij}$ = {0if $i < j$2if $i \geq j$δij = {02if i < jif i ≥ j

Choose the set of correct options

- [ ] $A$A is an upper triangular matrix.
- [ ] $A$A is a lower triangular matrix.
- [ ] $det(A)$ = 2det(A) = 2
- [ ] $det(A)$ = 8det(A) = 8

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$A$A is a lower triangular matrix.
$det(A)$ = 8det(A) = 8

Suppose $A$ = [4−13 2 0 1 3−20]A = [423−10−2310] and $C_{ij}$ Cij is the $(i, j)$(i, j)-th cofactor of the matrix $A$A.

Let $B$B be 3 ×× 3 matrix where $(i, j)$(i, j)-th entry is equal to $C_{ij}$ Cij. Find $det(B)$det(B).

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 49

*1 point*

Suppose $A$A and $B$B are two 3 ×× 3 matrices such that $det(A)$det(A) = 4 and $B$ = 3$A$B = 3A. Find the value of $\sqrt[3]{det(A^2 B)}$3 det(A2B)

√

.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 12

*1 point*

Check Answers

Your score is: 0/10